

Srihan Vege

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EDUCATION

Purdue University

B.S. in Computer Science & Math

West Lafayette, IN
Aug 2025 - Dec 2028

Relevant Courses: Linear Algebra, Programming I, Multivariable Calculus, Discrete Math, C Programming, Differential Equations, Statistics, Java Programming

EXPERIENCE

Bioqore

Software Engineering Intern

Boston, MA
June 2026 - Present

- Developed a synthetic biomanufacturing data framework in Python using SDV's Gaussian Copula and CTGAN models
 - Benchmarked across statistical fidelity, domain validity, utility, and privacy to identify the optimal generative model for USP upstream process data
 - Achieved 85% fidelity against real process data
- Designed a biological validity rule engine and model bake off harness that enforces domain plausibility constraints (titer/IVCD consistency, metabolite endpoint bounds) as hard gates
 - Flagged 93% of implausible synthetic records across 8 validation rules before model selection

NuStudio.AI

AI/ML Intern

Chicago, IL
May 2026 - Present

- Accomplished automated cross-language code migration, as measured by a 95% translation accuracy on test conversions, by engineering an LLM-powered agent under the Apex team
- Refined agent behavior, tested edge cases, and validated outputs across Cortava AI's credit union AI product suite
 - Improved tool reliability and surfacing failure modes across core platform workflows
- Designed and implemented REST endpoints for Sarvos's Action Broker, covering typed action proposal contracts and validate/execute flows across billing, document, and matter-intake domains
 - Enabled deterministic, auditable execution of agent proposed actions across all endpoint domains

Algoverse

CS/ML Engineer

Palo Alto, CA
Aug 2024 - Apr 2025

- Built a multi-turn LLM framework with prompts that help reduce accuracy degradation due to sycophancy by 43%
- Designed and implemented static and rationale based feedback pipelines to simulate human like interactions
- Co-first authored paper presented at NAACL SRW; 25+ citations, including work from Harvard and Google DeepMind

Northwestern University

Machine Learning Engineer - Bagci Lab

Chicago, IL
June 2024 - Feb 2025

- Built a CNN model that enables radiologists to accurately determine the peak enhancement point of an aortic scan
 - Predicted time of peak enhancement and actual time were off by a mode of 0.3 seconds, a 97% reduction from previous test results
 - Improved prediction accuracy by 50% on 272 CT scans, reducing error from 200 to 100 HU (measure of radiodensity in CT scans)
- Second-authored paper (submitted to IEEE EMBC) + patent pending

PROJECTS

Credit Card Fraud Detection

Python, Scikit-Learn, Pandas

Chicago, IL
May 2026

- Developed a machine learning pipeline for real-time credit card fraud detection using classification models and anomaly detection techniques achieving an 97.6% ROC AUC and an 86% F1-score on the fraud class (88% precision, 84% recall)
 - Processed and analyzed a highly imbalanced data of 284,807 transactions to improve fraud identification accuracy
 - Developed and compared multiple fraud detection models including Logistic Regression and XGBoost to optimize predictive performance

Vela - YC W26

TypeScript, GA4

San Francisco, CA
Jan 2026 - Feb 2026

- Built and deployed GA4 event tracking infrastructure across Vela's website to measure user workflow, conversion funnels, and retention

SKILLS

Frontend: React, Next.js, Tailwind, HTML, CSS, browser APIs

Backend: API design, Node.js, PyTorch, MongoDB, SQL

Languages: Python, TypeScript, Java, C, JavaScript